

Fourier-transform infrared spectroscopy (FTIR)

How does infrared spectroscopy work?

A beam of light in the infrared region of the electromagnetic spectrum is shined at a sample. Different types of bonds vibrate (by stretching, bending, wagging, etc.) at different wavelengths, which can be used to derive the types of bonds present in organic compounds.

How do you read an infrared spectroscopy graph?

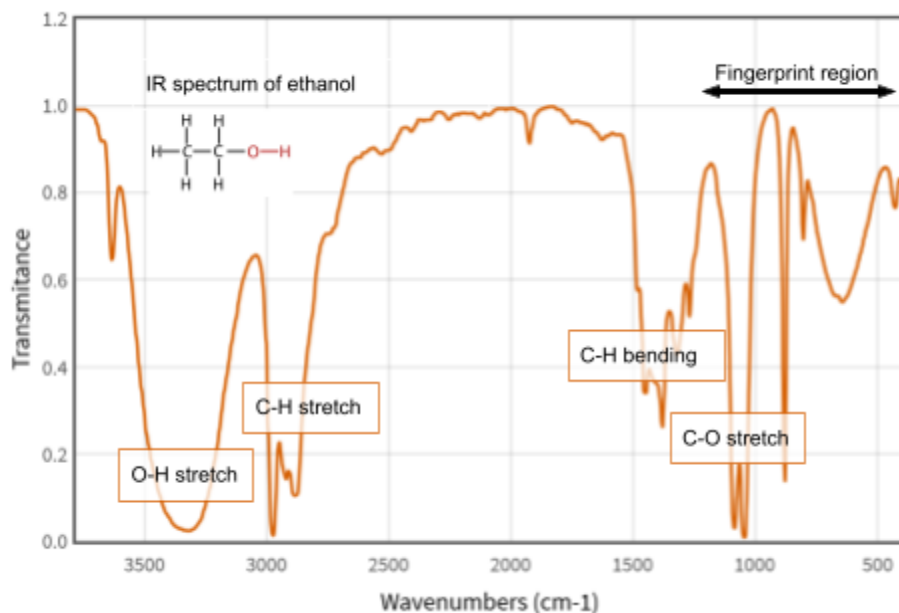


Table 1

Infrared absorption data

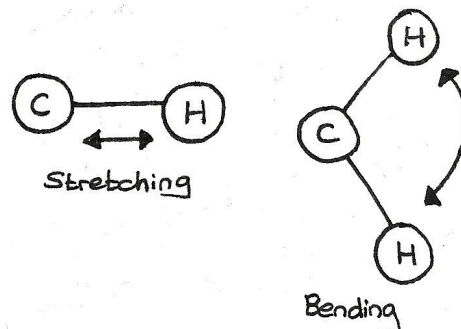
Bond	Wavenumber /cm ⁻¹
N-H (amines)	3300–3500
O-H (alcohols)	3230–3550
C-H	2850–3300
O-H (acids)	2500–3000
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C-O	1000–1300
C-C	750–1100

An IR spectroscopy graphs the sample's transmittance for each wavenumber. Wavenumbers are the reciprocal of the wavelength, i.e. $3000 \text{ cm}^{-1} = 3.33 \mu\text{m}$. Higher wavenumbers correspond to higher energy light being absorbed.

To determine a molecule's structure from its IR spectrum, determine the location of an absorption band (i.e. the dip at 3300 cm^{-1} above) and look at an infrared absorption data table. Looks like this corresponds to an O-H bond! Repeat for every notable dip on the spectrum and you'll find that the spectrum lines up with ethanol. There is a lot more nuance here, so I recommend reading up!

Where does the Fourier-transform part of the name come from?

When shining light at the sample, instead of iterating through each wavelength one by one (like UV-Vis spectrometers do), beams with different combinations of wavelengths are sequentially shined at the sample. Then, the computer takes the data and infers the absorptions at each wavelength using Fourier transform.



Instructions on using the Breakerspace FTIR Spectrometer

The Breakerspace has a Nicolet iS5 Fourier-transform infrared (FTIR) spectrometer, which is capable of analysing solids, liquids, gases, and thin films.



Instrument start-up

- Log on to instrument workstation using your MIT Kerberos
- Start OMNIC software
- Verify instrument connection via system status
- Remove protective cover from ATR crystal plate
- Clean ATR crystal
 - To clean the ATR crystal, wipe with isopropyl alcohol and a Kimwipe
- Collect background in the software

Operation

- Clean ATR crystal again if needed
- Place a sample over the ATR crystal
 - For *solid* samples
 - Ensure they are smooth and free of contaminants on the surface
 - For *powder* samples
 - Place enough powder to cover the ATR crystal completely.
 - Screw in either the concave or self-leveling pressure tip
 - Turn the black knob to push down on the sample until the clutch slips. The goal is to cover the surface of the ATR crystal completely!
 - For *liquid* samples:
 - Place a droplet on the ATR crystal.
 - Volatile samples should be covered with the volatiles cover.
- Collect sample in the software
- Save the sample spectrum, and repeat!

Instrument shut down:

- Save all data (each spectrum must be individually selected and saved)
- Close OMNIC and log-off the workstation
- Clean ATR crystal
- Put cover on crystal plate and clamp in place using pressure tower
- Do not switch the machine off (the manufacturer recommends leaving the instrument powered when not in use)

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!